

# MEI ROSA WIDYAWATI

Surabaya, Indonesia | +62 8814922923 | meirosa350@gmail.com  
linkedin.com/in/meirosa | github.com/meirosa | portofolio-mrosa.vercel.app

## Summary

---

An Applied Bachelor's student in Informatics Engineering at the Electronic Engineering Polytechnic Institute of Surabaya (PENS), currently awaiting graduation. Experienced in software development through internships and academic projects, including web and mobile application development, software testing, and system implementation. Familiar with Agile software development practices and experienced in collaborating within cross-functional development teams.

## Education

---

|                                                      |                      |
|------------------------------------------------------|----------------------|
| <b>Politeknik Elektronika Negeri Surabaya</b>        | 2025 – Expected 2026 |
| Applied Bachelor’s Degree in Informatics Engineering |                      |
| Current GPA: 3.78/4.00                               |                      |
| Awaiting Graduation Certificate (SKL)                |                      |
| <b>Politeknik Elektronika Negeri Surabaya</b>        | 2022 - 2025          |
| Associate Degree in Informatics Engineering          |                      |
| GPA: 3.59/4.00                                       |                      |

## Professional Experience

---

**Product Knowledge, Product Owner, Quality Assurance**  
**PT. Inovasi Teknologi Solusindo (Trustmedis) | Surabaya**  
**July 2024 - December 2024**

- Collaborated with Product Owners and development teams throughout Agile software development sprints.
- Conducted functional, integration, and regression testing for healthcare management systems.
- Identified, documented, and tracked software defects using Redmine.
- Assisted in product documentation, user acceptance testing (UAT), release validation, and quality improvement through cross-functional collaboration.

## Projects

---

**Comparative Evaluation of Fuzzy Inference System Models for Flood Vulnerability Assessment**  
**Final Project | Author & Full Stack Developer**  
**November 2025 – July 2026**

**Tech Stack:** Next.js, Python, FastAPI, Supabase, PostgreSQL, Leaflet, QGIS

- Designed and developed a web-based GIS application for flood vulnerability assessment using spatial open data.
- Implemented and evaluated Mamdani, Sugeno, and Tsukamoto Fuzzy Inference System models.
- Developed backend services and integrated them with a responsive frontend.

**Hotel Management System**

**Academic Project | Front-End Developer**

**May 2026**

**Tech Stack:** Laravel, PostgreSQL

- Collaborated with a development team to build a hotel management system.
- Developed responsive guest management interfaces using Laravel.
- Integrated frontend pages with backend data to ensure smooth user interactions.

**Flood Risk Mapping Visualization in Surabaya City Based on Open Map Data Using the Fuzzy Method**

**Final Project | Author & Full Stack Developer**

**July 2024 – June 2025**

**Tech Stack:** Next.js, Python, FastAPI, Supabase, PostgreSQL, Metabase, QGIS

- Developed a GIS-based flood vulnerability visualization system.
- Implemented the Fuzzy Mamdani method for flood risk assessment.
- Built backend services using Python and integrated them with Next.js.
- Visualized flood information through interactive dashboards and maps.

**Wallink**

**Academic Project | Front-End Developer & Quality Assurance**

**February 2024 – July 2024**

**Tech Stack:** Flutter, Figma, SQLite

- Collaborated in an Agile development team to build a mobile application.
- Developed responsive Flutter interfaces based on Figma designs.
- Performed functional testing and verified bug fixes to improve application quality.

**PensStore**

**Academic Project | Front-End Developer**

**February 2024 – June 2024**

**Tech Stack:** Flutter, Figma, SQLite

- Developed responsive mobile interfaces using Flutter.
- Implemented Figma designs into functional application screens.
- Collaborated with team members to improve application usability.

**Organizations & Leadership**

---

**Student Speaker | Informatics Engineering Program Socialization** 2026  
**Politeknik Elektronika Negeri Surabaya (PENS)**

- Represented the Applied Informatics Engineering program during outreach activities.
- Shared academic experiences and introduced the study program to prospective students.

**Member | Software Development Community** 2022 – Present

- Collaborated on software development projects and participated in technical workshops.
- Expanded knowledge of software engineering practices and emerging technologies.

**Event Division | LKMM-TD PENS** 2024

- Planned and coordinated event activities and logistics.
- Managed schedules and participant engagement during the program.

**Secretary | DKR Badegan** 2022 – 2023

- Managed organizational administration, documentation, and official correspondence.

- Prepared meeting minutes, reports, and administrative documents.
- Coordinated communication between members and organizational leaders.

#### **Liaison Officer | Dynamic HIMIT PENS**

2023

- Served as the primary liaison between the organizing committee and external participants.
- Coordinated communication to ensure smooth event execution.

#### **Achievements**

---

##### **Gold Medal | Physics Olympiad | Province Level**

CV. Divya Cahaya Prestasi, 2021

##### **Bronze Medal | Mathematics Olympiad | National Level**

CV. Divya Cahaya Prestasi, 2021

##### **Gold Medal | English Olympiad | National Level**

CV. Divya Cahaya Prestasi, 2021

##### **Top 7 | Essay Competition | National Level**

Universitas Gunung Jati, 2024

Authored the essay "Sustainable Agricultural Sector Development as a Creative Approach to Supporting the Sustainable Development Goals (SDGs)"

#### **Skills**

---

**Programming Languages:** Python, PHP, JavaScript, TypeScript, Dart, HTML5, CSS3, SQL

**Frameworks & Libraries:** Next.js, React, Laravel, Flutter, FastAPI, Vue.js, Tailwind CSS, Bootstrap

**Databases:** PostgreSQL, MySQL, SQLite, Supabase

**Tools & Platforms:** Git, GitHub, Postman, Redmine, Figma, Notion, Google Docs, Markdown, Visual Studio Code

**GIS & Data Visualization:** Leaflet, QGIS, Metabase, Grafana

**Methodologies:** Agile, Scrum

**Languages:** Indonesian (Native), English (Intermediate)